Andrea D'Angelo

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Short Bio



Andrea D'Angelo is a Postdoctoral researcher at Aarhus University and, from November 2026, a Danish Data Science Academy (DDSA) Postdoctoral Fellow with his own funded project on *Robust Graph Unlearning*. He was awarded a Ph.D. cum laude with the Doctor Europaeus distinction by the University of L'Aquila in 2026. His research focuses on Trustworthy Artificial Intelligence and Machine Unlearning. He has authored more than 15 papers in leading venues, including NeurIPS, EMNLP, IJCAI, CIKM, and the Machine Learning journal. He has organized international workshops, delivered a tutorial at ECML-PKDD 2025, served as Proceedings Chair for CIKM 2024 and WSDM 2026, and given invited talks at leading institutions, including the Technical University of Munich (TUM).

Research Experience

Nov 2026–	**DDSA Postdoc**, Datalogi, Aarhus University. Project: *Robust Graph Unlearning*. Funded by the Danish Data Science Academy.
May–Oct 2026	**Postdoctoral Researcher**, Datalogi, Aarhus University, (Host: Prof. Davide Mottin).
Dec 2025–Apr 2026	**Research Scholarship**, University of L'Aquila, industry collaboration with Apprendo. Apprendo research group member since 2026.
2025	**Visiting Ph.D. Student** (6 months), Datalogi, Aarhus University. Graph Unlearning and Differential Privacy.
2022–26	**Ph.D. Student** (XXXVIII cycle), University of L'Aquila.
2021–22	**Research Scholarship** (9 months), University of L'Aquila, *Territori Aperti* project. LLM-based Dense Retrieval.

Education

2022–26	**Ph.D.**, Computer Science, University of L'Aquila, cum Laude, Doctor Europaeus. Thesis: *Trustworthy AI Through Principled Advances in Machine Unlearning, Algorithmic Bias Mitigation, and Explainability*.
2020–22	**M.Sc.**, Computer Science (LM-18), University of L'Aquila, 110/110 cum Laude, 22/10/2022. Thesis: *Clouds of Vectors for Density-Based Information Retrieval*.
2016–19	**B.Sc.**, Computer Science (L-31), University of L'Aquila, 110/110 cum Laude, 21/10/2019. Thesis: *Application of Standard Compilation Techniques to Non-Linear Languages*.

Honours & Awards

2026	**DDSA Postdoc Grant** (Danish Data Science Academy) for the project *Robust Graph Unlearning* (1,365,000 DKK / ~€183,000).
2024	**EUGAIN Virtual Mobility Grant**, COST Action CA19122 (E-COST-GRANT-CA19122-04419a60).

Teaching

LUISS University of Rome

2026	Lecturer, *Big Data Management: Data Mining and Analysis for Management Decisions*, 21 hours.

Engage.EU Summer School (LUISS University of Rome)

2026	Lecturer, *Privacy Preservation for Machine Learning*, 2 hours.

University of L'Aquila

2024	Co-creator, MOC course *Reinforcement Learning*, SoBigData Academy[1].
2023, 2024	Teaching Assistant, *Deep Neural Networks Laboratory* (M.Sc. Computer Science, LM-18), 8 hours.
2023, 2024	Teaching Assistant, *Reinforcement Learning Laboratory*, Artificial Intelligence course (M.Sc. Computer Science, LM-18), 5 hours.

Research Software

Lethe

Lethe is an open-source benchmark for Edge Unlearning in Graph Neural Networks. It provides a standardized suite of link inference attacks for evaluating edge unlearning methods on structurally informative graphs. Accepted at **NeurIPS 2026**, Evaluations and Datasets Track.

ERASURE

ERASURE is an open-source Python framework designed to make Machine Unlearning research reproducible and comparable. It provides a configuration driven interface for running and benchmarking forgetting algorithms across multiple models, datasets, and evaluation metrics. Published as a resource paper at **CIKM 2025** and demonstrated at **IJCAI 2025**.

[1] https://sobigdata.unipi.it/course/view.php?id=17

Supervision

Scholars

2025	Francesca Ciccarelli, Ph.D. Student, University of L'Aquila. Machine Unlearning applied to LLMs.
2025	Manuel Antonio Vilas Valiente, Visiting Scholar, University of L'Aquila. Machine Unlearning applied to Time Series.

Theses

2026	Susanna Peretti, M.Sc. Computer Science, University of Urbino. *Machine Unlearning applied to Time Series and Human Activity Recognition*.
2025	Fabio D'Andreamatteo, M.Sc. Data Science, University of L'Aquila. *HybridKAN: Leveraging Multi-Sized Sub-MLPs for Enhanced Performance*.
2025	Samuele Ramassone, M.Sc. Computer Science, University of L'Aquila. *Beyond the Systematic Evaluation of Large Language Models: From Theory to Practical Analysis of Results*.
2025	Gabriele Tagliente, M.Sc. Computer Science, University of L'Aquila. *From ERASURE to Evaluation of Machine Unlearning Techniques*.
2025	Matteo Iezzi, M.Sc. Data Science, University of L'Aquila. *Recent Advances in Chain-of-Thought Prompting: Current Trends and Automation Approaches*.

Outreach

2024, 2025	Tutor, Pinkamp STEM outreach program.

Service

Organization

2026	**Co-organizer** of Parallel Session "Knowledge Graphs: Connecting Theory, Algorithms, and Applications", D3A 4.0 (Danish Digitalization, Data Science and AI), Nyborg Strand, Denmark.
2026	**Co-organizer** of Workshop on Innovations, Privacy-preservation, and Evaluations of machine Unlearning Techniques (WIPE-OUT 2), ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases) 2026, Naples.
2025	**Co-organizer** of Workshop on Web & Graphs, Responsible Intelligence, and Social Media (WEB&GRAPH 2026), WSDM (The 19th ACM International Conference on Web Search and Data Mining) 2026, Boise, Idaho, USA.
2025	**Proceedings Chair** of WSDM (ACM International Conference on Web Search and Data Mining) 2026.

2025	**Co-organizer** of Workshop on Innovations, Privacy-preservation, and Evaluations of machine Unlearning Techniques (WIPE-OUT 2025), ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases) 2025, Porto.
2025	**Program Committee Chair** of Workshop on Innovations, Privacy-preservation, and Evaluations of machine Unlearning Techniques (WIPE-OUT 2025), ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases) 2025, Porto.
2025	**Co-organizer** of Tutorial: Unraveling Graph Counterfactual Explainability: from Theoretical Foundations to Technical Mastery, ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases) 2025, Porto.
2024	**Proceedings Chair** of CIKM (ACM International Conference on Information and Knowledge Management) 2024.
2024	**Co-organizer** of "Territori Aperti: dai risultati alle sfide future" event.

Member of the Program Committee

2026	Springer Artificial Intelligence Review Journal
2026	Automated Software Engineering Journal
2027	AAAI (AAAI Conference on Artificial Intelligence).
2026	ITADATA (Italian Conference on Big Data and Data Science).
2025	ACM Web Conference (formerly known as the International World Wide Web Conference)
2025, 2026	KDD (ACM SIGKDD Conference on Knowledge Discovery and Data Mining) Datasets & Benchmarks.
2024, 2025, 2026	CIKM (ACM International Conference on Information and Knowledge Management). **Distinguished Reviewer** in 2024.
2025, 2026	ECML-PKDD (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases).
2025	ICPE Data Challenge (International Conference on Performance Engineering).
2024, 2026	LL4MSE Workshop @ STAF (Software Technologies: Applications and Foundations).
2024	DELTA Workshop @ KDD 2024 (ACM SIGKDD Conference on Knowledge Discovery and Data Mining).
2024	MIDAS Workshop @ ECML-PKDD 2024 (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases).
2023	BIAS Workshop @ ECIR 2023 (European Conference on Information Retrieval).

Presentations

Invited Talks

IT1. **D'Angelo**, **A.** *Trustworthy AI and the Right to be Forgotten: The Role of Machine Unlearning.* LUISS Guido Carli. July 2026.

IT2. **D'Angelo**, **A.** *Perspectives on Machine Unlearning and Privacy-preserving Machine Learning.* Technical University of Dresden. Feb. 2026.

IT3. **D'Angelo**, **A.** *Machine Unlearning: Safeguarding Privacy and Security in AI Models.* Technical University of Munich. July 2024.

IT4. **D'Angelo**, **A.** *Privacy Implications in the Training and Use of Large Language Models.* Territori Aperti, Napoli. 2024.

IT5. **D'Angelo**, **A.** *Uncovering Gender Bias in Academia.* AIGAP 2023 (Algorithmic Biases in Artificial Intelligence from Interdisciplinary Perspectives), L'Aquila. 2023.

Talks

T1. **D'Angelo**, **A.** *Forgetting Right: Towards Reproducible Benchmarks in Machine Unlearning with ERASURE.* WIPE-OUT Workshop @ ECML-PKDD 2025 (European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases), Porto. 2025.

T2. **D'Angelo**, **A.** *HybridKAN: Leveraging Multi-Sized Sub-MLPs for Enhanced Performance.* ITADATA (Italian Conference on Big Data and Data Science) 2025, Turin. 2025.

T3. **D'Angelo**, **A.** *A Novel Relevance Score for Unsupervised Retrieval with Large Language Models.* KDD 2024 (ACM SIGKDD Conference on Knowledge Discovery and Data Mining), Barcelona. 2024.

T4. **D'Angelo**, **A.** *Echocardiographic Epicardial Adipose Tissue Quantification: Challenges and Insights.* PerFail Workshop @ PerCom (International Conference on Pervasive Computing and Communications) 2024, Biarritz. 2024.

T5. **D'Angelo**, **A.** *Data-Driven Analysis of Gender Fairness in the Software Engineering Academic Landscape.* ECSA 2023 (European Conference on Software Architecture), Istanbul. 2023.

Posters

P1. **D'Angelo**, **A.** & Mottin, D. *Robust Graph Unlearning.* D3A 4.0 (Danish Digitalization, Data Science and AI), Nyborg Strand, Denmark, 2026.

P2. Ciccarelli, F., **D'Angelo**, **A.** & Stilo, G. *Towards a Novel Visual Evaluation of Algorithmic Bias: Insights on the Italian Academic System.* KDD (ACM SIGKDD Conference on Knowledge Discovery and Data Mining), 2024.

P3. **D'Angelo**, **A.**, Di Marco, A. & Stilo, G. *A Novel Relevance Score for Unsupervised Retrieval with Large Language Models.* KDD (ACM SIGKDD Conference on Knowledge Discovery and Data Mining) 2024, Barcelona, Spain, 2024.

Tutorials

D1. Prenkaj, B., Zaradoukas, E., Stilo Giovanni andD'Angelo, A. & Prado-Romero, M. *Unraveling Graph Counterfactual Explainability: from Theoretical Foundations to Technical Mastery*. ECML-PKDD 2025, Porto, Portugal. Sept. 2025.

Publications

Google Scholar: https://scholar.google.com/citations?user=9WIwwWwAAAAJ

Peer-reviewed Conference Papers

C1. Bushipaka Praveen andD'Angelo, A., Passaro, L. & Cucinotta, T. *GRACE: Gradient-guided Coreset Selection for LLM Unlearning* in *Findings of the Association for Computational Linguistics: **EMNLP 2026*** (2026).

C2. **D'Angelo**, **A.**, Gullo, F., Mottin, D., Ranu, S. & Stilo, G. *Lethe: Link Inference Attacks For Evaluation of Edge Unlearning Methods* in *Advances in Neural Information Processing Systems (**NeurIPS 2026**), Evaluations and Datasets Track* (2026).

C3. **D'Angelo**, **A.**, Savelli, C., Tagliente, G., Giobergia, F., Baralis, E. & Stilo, G. *How to Make Reproducible Research in Machine Unlearning with ERASURE* in *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*, ***IJCAI-25*** Demo Track (2025), 11025–11029.

C4. D'Andreamatteo Fabio andD'Angelo, A. & Stilo, G. *HybridKAN: Leveraging Multi-Sized Sub-MLPs for Enhanced Performance* in *ITADATA (Italian Conference on Big Data and Data Science) 2025* (2025).

C5. **D'Angelo**, **A.**, Savelli, C., Tagliente, G., Giobergia, F., Baralis, E. & Stilo, G. *ERASURE: a Modular and Extensible Framework for Machine Unlearning* in *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (**CIKM 2025**)* (2025).

C6. Di Sipio Claudio andD'Angelo, A., Rubei, R. & Politowski, C. *On the Need for Reproducibility Guidelines for Open-Source Games: A itch.io Case Study* in *2025 IEEE Conference on Games (CoG)* (2025), 1–8.

C7. d'Aloisio Giordano andD'Angelo, A., Marzi, F., Di Marco, D., Stilo, G. & Di Marco, A. *Data-Driven Analysis of Gender Fairness in the Software Engineering Academic Landscape* in *Software Architecture. ECSA 2023 Tracks, Workshops, and Doctoral Symposium* (2024), 89–103.

C8. **D'Angelo**, **A.** & d'Aloisio, G. *Grammar-Based Anomaly Detection of Microservice Systems Execution Traces* in *Companion of the 15th ACM/SPEC International Conference on Performance Engineering* (2024), 77–81.
Best Short Paper Award (ICPE 2024).

C9. **D'Angelo**, **A.**, Di Sipio, C., Politowski, C. & Rubei, R. *PlayMyData: a curated dataset of multi-platform video games* in *Proceedings of the 21st International Conference on Mining Software Repositories* (2024), 525–529.

C10. Bianchi, A., d'Aloisio, G., **D'Angelo**, **A.**, Di Marco, A., Di Matteo, A., Leone, J., Scoccia, G., Stilo, G. & Traini, L. *DIORAMA: Digital twIn fOR sustAinable territorial MAnagement* in *Proceedings of itaDATA (Italian Conference on Big Data and Data Science)* (2022), 144–155.

C11. d'Aloisio, G., Stilo, G. & Di Marco Antinisca andD'Angelo, A. *Enhancing Fairness in Classification Tasks with Multiple Variables: A Data- and Model-Agnostic Approach* in *Advances in Bias and Fairness in Information Retrieval* (2022), 117–129.

Journal Articles

J1. **D'Angelo**, **A.**, Savelli, C., Giobergia, F., Baralis, E. & Stilo, G. On the Evaluation of Machine Unlearning Methods: A Multi-domain Classification Benchmark. ***Machine Learning*** **115,** 161 (2026).

J2. **D'Angelo**, **A.**, Gullo, F. & Stilo, G. The Forget-Set Identification Problem. ***Machine Learning*** **114** (2025).

J3. Bianchi, A., Zelli, V., **D'Angelo**, **A.**, Di Matteo, A., Scoccia, G., Cannita, K., Dimas, A. S., Glentis, S., Zazzeroni, F., Alesse, E., Di Marco, A. & Tessitore, A. A method to comprehensively identify germline SNVs, INDELs and CNVs from whole exome sequencing data of BRCA1/2 negative breast cancer patients. ***NAR Genomics and Bioinformatics*** **6,** lqae033 (2024).

J4. **D'Angelo**, **A.**, d'Aloisio, G., Marzi, F., Di Marco, A. & Stilo, G. Uncovering gender gap in academia: A comprehensive analysis within the software engineering community. ***Journal of Systems and Software*** **217,** 112162 (2024).

J5. d'Aloisio, G., **D'Angelo**, **A.**, Di Marco, A. & Stilo, G. Debiaser for Multiple Variables to enhance fairness in classification tasks. ***Information Processing & Management*** **60,** 103226 (2023).

J6. Della Penna, G., Orefice, S. & **D'Angelo**, **A.** Exploiting spatial relations for grammar-based specification of multidimensional languages. ***Knowledge and Information Systems*** **65,** 3995–4020 (2023).

Book Chapters

B1. Wilson Takaoka, A. J., Alboaie, L., Cutrupi, C. M., **D'Angelo**, **A.**, Di Marco, A. & Hillston, J. in *Actions for Gender Balance in Informatics Across Europe* 271–297 (2025).

Other Grants & Recognitions

2025	**ERASMUS Traineeship Grant** for European mobility.
2024	**Distinguished Reviewer Award**, CIKM 2024.
2024	**Mobility Grant**, KDD 2024.
2024	**Best Short Paper Award**, ICPE 2024.
2022	**Ph.D. Scholarship** (publicly funded), University of L'Aquila.

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